

Enhanced Efficient Single Image Super-Resolution (CRAFT) using Transformer Optimisation

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Abstract—Transformer-based approaches have shown significant promise in Single Image Super-Resolution (SISR) due to their ability to model long-range dependencies. However, many existing methods focus primarily on enhancing global feature extraction through attention mechanisms, often neglecting the role of high-frequency cues that are vital for restoring fine image details. Through extensive analysis, they observed that while transformers excel at capturing low-frequency information, they often fall short in recovering high-frequency textures compared to convolution-based networks. To bridge this gap, they propose a set of targeted architectural upgrades that culminate in a unified framework called the Cross-Refinement Adaptive Feature Modulation Transformer (CRAFT). Our improvements include: a dynamic window shifting mechanism within the Shifted Rectangle Window Attention Block (SRWAB) for spatially adaptive attention, a Cross-Channel High-Frequency Enhancement (CCHFE) module embedded in the High-Frequency Enhancement Residual Block (HFERB) to emphasize texture-rich features, and a Gated Dual-Path Fusion strategy in the Residual Cross-Refinement Fusion Group (RCRFG) to effectively integrate low- and high-frequency components. Furthermore, a frequency-aware post-training quantization (PTQ) method that improves model efficiency through adaptive clipping and refined boundary constraints is introduced. This PTQ method is also adaptable to other transformer-based SISR architectures. The experimental results confirm that the improved CRAFT model delivers competitive or superior performance under both full-precision and quantised conditions, highlighting the benefits of our frequency-informed, hybrid design.

Index Terms—Single Image Super-Resolution, Optimisation, Super-Resolution, transformer, frequency priors, CNN Super-Resolution.

I. INTRODUCTION

Single Image Super-Resolution (SISR) has garnered significant attention in recent years due to its promising applications across various domains, including surveillance video enhancement and medical imaging [1], [2], restoration of old photographs [3], [4], and efficient image transmission [5]. Despite its widespread applicability, SISR remains a fundamentally challenging problem: reconstructing a high-resolution (HR) image from its low-resolution (LR) counterpart is an ill-posed task, as multiple plausible HR solutions may correspond to the same LR input.

Recent advances in deep learning have led to significant breakthroughs in SISR [6]–[8], leveraging the expressive capabilities of neural networks. Notably, convolutional neural networks (CNNs) with innovative architectural designs—such

as residual and dense connections—have demonstrated substantial improvements in image reconstruction accuracy [9], [10]. Furthermore, the incorporation of attention mechanisms, which adeptly capture both spatial and channel-wise relationships, has propelled performance even further [11]–[14].

More recently, the introduction of Transformer architectures has marked a pivotal shift in SISR research. Their ability to model long-range dependencies has enabled state-of-the-art performance across several benchmarks [3], [15]–[18]. However, several key limitations persist. First, while many contemporary models have successfully integrated advanced modules—such as global attention in Transformers or sophisticated convolutional structures—they often neglect the role of frequency-domain information, which can be crucial for accurate image reconstruction. They argue that a thorough investigation of frequency priors may offer valuable insights into model behavior and performance.

Second, the high computational demands of Transformer-based SISR models remain a barrier to practical deployment, especially in resource-constrained environments. Thus, both architectural design and computational efficiency must be jointly addressed to realize the full potential of SISR in real-world applications.

A. Contributions

To address the limitations of current transformer-based SISR methods in handling high-frequency details and computational efficiency, they introduce three key architectural enhancements to the baseline CRAFT framework:

- 1) **Shifted Rectangle Window Attention Block (SRWAB) with Dynamic Windowing:** they improve the original SRWAB by incorporating a dynamic window shifting mechanism. This allows the attention windows to adapt spatially across different layers, enabling better modeling of cross-region dependencies and reducing blind spots in attention computation.
- 2) **Cross-Channel High-Frequency Enhancement (CCHFE) Module in HFERB:** To better extract and preserve fine textures, they integrate a CCHFE module into the High-Frequency Enhancement Residual Block (HFERB). This module enhances frequency-specific feature maps across channels using grouped convolutions and channel recalibration, thereby

reinforcing high-frequency details often lost in transformer layers.

- 3) **Gated Dual-Path Fusion in RCRFG:** they introduce a gated dual-path fusion mechanism in the Residual Cross-Refinement Fusion Group (RCRFG), which separately processes low- and high-frequency features before selectively integrating them using a learnable gating function. This design facilitates refined feature modulation by balancing coarse contextual information and sharp edge details.

B. Organization

The remainder of the article is outlined as follows. In Section II, the background knowledge is introduced. Section III elaborates the proposed method and Section IV presents the dataset description. The results are reported and analyzed in Section V. Section VI concludes the article and points out future work.

II. BACKGROUND AND RELATED WORKS

A. CNN Based Models

Since the pioneering work SRCNN [6] has achieved significant progress in SISR, various CNN-based works have been proposed. Kim et al. [27] presented an SR method using deep networks by cascading 20 layers, demonstrating promising results. Building upon this, Lim et al. [20] introduced the enhanced deep super-resolution (EDSR) network, which achieved a significant performance boost by removing the batch normalization layer [28] from the residual block and incorporating additional convolution layers. Ahn et al. [29] designed an architecture with an increased number of residual blocks and dense connections, further improving the SR performance. In pursuit of lightweight models, Hui et al. [30] proposed a selective fusion approach, employing cascaded information multi-distillation blocks to construct an efficient model. Li et al. [31] introduced a method involving predefined filters and used a CNN to learn coefficients, which were then linearly combined to obtain the final results. Sun et al. [32] proposed a hybrid pixelunshuffled network (HPUN) by introducing an efficient and effective downsampling module into the SR task.

B. Transformer based Models

Liang et al. [3] proposed SwinIR, a robust baseline model for image restoration, leveraging the Swin Transformer [33]. CAT [15] modified the window shape and introduced a rectangle window attention to obtaining better performance. Chen et al. [16] proposed a pre-trained image processing transformer and showed that pre-trained mechanism could significantly improve the performance for low-level tasks. Li et al. [17] comprehensively analyzed the effect of pretraining and proposed a versatile model to tackle different low-level tasks. Lu et al. [18] proposed a lightweight transformer to capture long-range dependencies between similar patches in an image with the help of the specially designed efficient

attention mechanism. Zhang et al. [34] introduced a shift convolution and a group-wise multiscale self-attention to reduce the complexity of transformers. HAT [35] introduced a hybrid attention mechanism to enhance the performance of window-based transformers.

C. Quantisation in SISR

Quantization plays a critical role in compressing and speeding up models, primarily divided into Quantization-aware Training (QAT) and Post-Training Quantization (PTQ). QAT fine-tunes weights and activations during training, which can enhance model performance despite requiring more training time. Ma et al. [36], explored the potential of weight binarization in SR, demonstrating the possibility of extreme compression.

In contrast to QAT, PTQ offers a fast and efficient way to quantization, only needing a small set of calibration data. This method is especially useful when there is limited access to the complete dataset or when quick model deployment is needed. PTQ4SR [26] uses a two-step quantization approach specifically designed for the dynamic requirements of SR tasks. Yet, there is a notable gap in quantization methods, especially for transformer-based SR models. Our PTQ method can be expanded to a general quantization strategy for transformer-based SISR methods, promoting the development of more efficient and scalable SR solutions

D. Frequency Analysis

To thoroughly investigate how different SISR models handle frequency information, the authors conduct a physically realistic frequency-domain analysis across both CNN- and transformer-based architectures. The goal is to understand the sensitivity of these models—especially transformers—to the loss of high-frequency components, which are critical for preserving texture and fine image details.

1) Frequency-Dropping Procedure: Given a high-resolution image X^{HR} , the authors first apply a 2D Fast Fourier Transform (FFT) to convert it into the frequency domain. The FFT output is spatially organized such that low-frequency components are located at the center and high-frequency components extend outward radially.

The frequency elements are sorted by their Euclidean distance from the center of the spectrum. Then, for a given drop ratio $\gamma \in [0, 1]$, the furthest $\gamma \cdot L$ frequency components (where L is the total number of frequency elements) are zeroed out. The modified frequency spectrum is then converted back to the spatial domain using an inverse FFT (IFFT):

$$X_{Drop(\gamma)}^{HR} = \text{IFFT}(\text{Drop}(\mathcal{F}(X^{HR}))) \quad (1)$$

where $\mathcal{F}(\cdot)$ denotes the FFT operation, and $\text{Drop}(\cdot)$ sets the top $\gamma \cdot L$ components to zero.

The frequency-dropped high-resolution image is then downsampled using bicubic interpolation:

$$X_{Drop(\gamma)}^{LR} = \text{Downsample}(X_{Drop(\gamma)}^{HR})$$

This low-resolution image is fed into various SISR models to produce the super-resolved image $\hat{X}_{Drop(\gamma)}^{SR}$.

2) *Performance Metric: PSNR Drop Ratio*: To evaluate the impact of high-frequency removal on reconstruction quality, the authors define the PSNR drop ratio $R_{PSNR}^{Drop}(\gamma)$:

$$R_{PSNR}^{Drop}(\gamma) = \frac{P(0) - P(\gamma)}{P(0)} \quad (2)$$

where:

- $P(0)$ is the PSNR when no frequencies are dropped,
- $P(\gamma)$ is the PSNR after dropping the top $\gamma \cdot L$ high-frequency components.

This metric reflects the relative sensitivity of a model to frequency loss. Empirical results show that transformer-based models exhibit a significantly larger PSNR drop compared to CNN-based models, indicating greater dependence on high-frequency information.

3) *Relative Energy Drop Ratio*: To further interpret the PSNR decline, the authors analyze the proportion of total frequency energy removed. The relative energy drop ratio is defined as:

$$R_E^{Drop}(\gamma) = \frac{E^f(0) - E^f(\gamma)}{E^f(0)} \quad (3)$$

where:

- $E^f(0)$ is the total energy (sum of squared magnitude) of the original frequency spectrum,
- $E^f(\gamma)$ is the energy of the frequency spectrum after dropping $\gamma \cdot L$ components.

Plotting $R_{PSNR}^{Drop}(\gamma)$ against $R_E^{Drop}(\gamma)$ reveals that transformer models suffer more severe performance degradation for a given amount of energy loss compared to CNNs.

4) *Real-World Simulation: Mean Filter Blurring*: To simulate practical degradation scenarios such as optical blur or motion blur, the authors apply a mean filter to X^{HR} prior to downsampling. The result is:

$$X_{blurred}^{LR} = \text{Downsample}(\text{Blur}(X^{HR}))$$

They find that as the mean filter kernel size increases, PSNR deteriorates more rapidly in transformer models than in CNNs. Visual results also show that transformers introduce excessive blurring and structure loss in the output images under this setup.

5) *Key Observations*:

- **Transformer models** are more sensitive to the removal of high-frequency information, even when only a small portion of frequency energy is removed.
- **CNN models** demonstrate greater robustness to high-frequency loss and perform better in realistic degradation settings.

- These insights motivate a hybrid design, where CNN-generated high-frequency priors can support transformer-based models in recovering fine textures.

These foundational findings drive the development of the authors' proposed method, which integrates frequency-aware enhancements into the SISR pipeline.

III. METHODOLOGY - CRAFT MODEL

The CRAFT model (Convolutional Residual Attention Frequency Transformer) introduces a novel architecture designed to optimize the trade-off between efficiency and performance in single image super-resolution (SISR). At its core, CRAFT synergizes convolutional priors and transformer-based attention mechanisms, inspired by frequency-based analysis, to enhance both reconstruction accuracy and computational cost-effectiveness.

A. Overview of CRAFT Architecture

The CRAFT framework consists of a hierarchical structure with three primary stages:

- **Shallow Feature Extraction**
- **Deep Feature Extraction via Cross-Refinement Fusion Groups (CRFG)**
- **Image Reconstruction**

This structured flow allows CRAFT to encode fine-grained spatial and frequency features while remaining lightweight.

B. Shallow Feature Extraction

The process begins with a convolutional layer that extracts low-level features from the low-resolution input image I_{LR} . This layer outputs a feature map F_0 , which captures basic edge and texture details:

$$F_0 = H_{SF}(I_{LR})$$

where H_{SF} is the shallow feature extractor, typically a 3×3 convolution.

C. Deep Feature Extraction via CRFG Modules

The core of the CRAFT model lies in its **Cross-Refinement Fusion Groups (CRFG)**. A series of CRFG blocks are used to hierarchically extract and refine deep features. Each CRFG incorporates the following key components:

1) Residual Cross-Refinement Fusion Block (RCRFB):

Each CRFG contains multiple RCRFBs, which integrate local and global information efficiently. These blocks consist of:

- **High-Frequency Enhancement Residual Block (HFERB)**: Emphasizes high-frequency features essential for image sharpness.
- **Shifted Window Attention Block (SRWAB)**: Enhances the receptive field and context awareness using a local self-attention mechanism.
- **Cross-Refinement Fusion Block (CRFB)**: Facilitates cross-layer communication and feature blending.

This configuration ensures both frequency-aware detail enhancement and long-range dependency modeling.

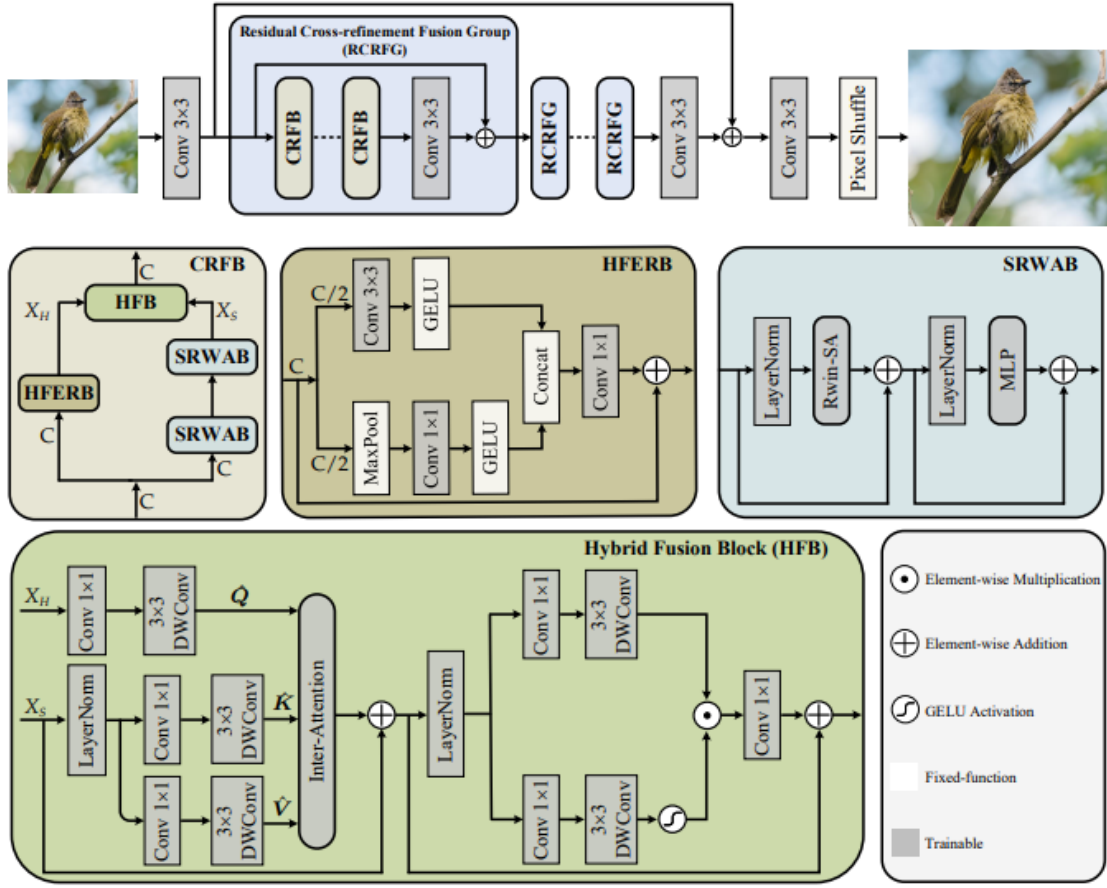


Fig. 1: Framework of CRAFT. HFERB extracts the high-frequency information from the input features, SRWAB captures the long-range dependency of input features, and HFB integrates the output of HFERB and SRWAB to cross refine the global features.

2) High-Frequency Enhancement Residual Block:

- To handle and construct high frequency data, an image feature representation is used for the HFERB.
- It consists of multiple Convolutional and Max-Pooling blocks.
- It comprises the local feature extraction (LFE) branch and the high-frequency enhancement (HFE) branch.
- LFE using 3×3 Convolution block and GELU.
- HFE using max pool, 1×1 convolution and GELU.
- Concatination and use of Skip Connection for stability and gradient.

3) Shifted Window Attention Block:

- To handle long range dependencies.
- Input divided into rectangles of $H \times W \times \frac{1}{r_h \times r_w}$
- It comprises query, key and value

$$Q_i = X_i W, \quad K_i = X_i W_K, \quad V_i = X_i W_V$$

for each i th rectangle

- Self Attention as:

$$\text{Attention}(Q_i, K_i, V_i) = \text{Softmax} \left(\frac{Q_i K_i^T}{\sqrt{d}} + B \right) V_i$$

where B is the dynamic relative position encoding.

- To capture information from different axes, use two types of rectangle windows: horizontal and vertical windows.
- Split the attention heads into two equal groups and compute the self-attention within each group separately

4) Cross-Refinement Fusion Block:

- To better integrate the merits of CNN and transformer (HFERB and SRWAB), a hybrid fusion block (HFB) is used.
- Formulate the output of HFERB as the high-frequency prior query and the output of SRWAB as key, value and calculate the inter-attention to refine the global features which are obtained from SRWAB.
- To overcome loss of information through the channel an inter-attention based on the channel dimension to explore channel dependencies is used.
- A 1×1 convolutional layer followed by a 3×3 depth-wise convolutional layer to generate the high-frequency query $Q \in \mathbb{R}^{H \times W \times C}$ based on the output of HFERB
- As to the output of SRWAB normalize the features by the LayerNorm layer and then use the same operation as

the query Q to get the key $K \in \mathbb{R}^{H \times W \times C}$ and the value $V \in \mathbb{R}^{H \times W \times C}$.

- Inter-Attention as
 $\text{Attention}(\hat{Q}, \hat{K}, \hat{V}) = \text{Softmax}\left(\frac{\hat{Q}\hat{K}^T}{\alpha}\right)\hat{V}$

D. Image Reconstruction

The final reconstruction is achieved using a pixel-shuffle upsampling module applied to the deep features F_D :

$$I_{SR} = H_{UP}(F_D + F_0)$$

The skip connection between the shallow and deep features ensures preservation of original structures and contributes to better gradient flow during training.

E. Efficiency Strategies

CRAFT employs several strategies to enhance efficiency:

- **Frequency-Inspired Optimization:** HFERB selectively enhances high-frequency components which are often lost in traditional SR models.
- **Lightweight Transformer Blocks:** SRWAB uses local attention and windowing techniques to reduce complexity compared to global self-attention.
- **Residual Connections and Cross-Fusion:** CRFB enables multi-path refinement and effective reuse of feature representations.

F. Loss Function

The model is trained using a combination of pixel-wise loss (L1 or L2) and potentially perceptual losses to balance sharpness and fidelity. While the exact loss function configuration may vary, the main objective remains minimizing the reconstruction error between I_{SR} and the ground truth high-resolution image I_{HR} .

Summary

CRAFT introduces a frequency-aware, attention-augmented transformer architecture that leverages both convolutional and transformer paradigms. By integrating components like HFERB, SRWAB, and CRFB into CRFG modules, it achieves a compelling balance between super-resolution quality and computational efficiency. This makes CRAFT particularly suitable for real-time and resource-constrained applications without sacrificing perceptual detail or sharpness.

IV. CONSIDERED IMPROVEMENTS

- **Adaptive SRWAB:** Prediction of window size for each small patch and attention mechanism after matching feature maps using interpolation (bilinear or trilinear)
- **Adaptive SRWAB:** Dynamic Window Size Selection for the entire image using softmax probability (predictive architecture before SRWAB)
- **Edge Aware CNN Modules** - Laplacian and MSD Convolution
- **Progressive Refinement Across Layers** using transformer blocks (capture the image and then recreate the

image feature map) to create all low freq features first and then use the final low freq map for Cross-Attention fusion (Use Low freq as **Q** and CNN output as **K**)

- **Channel Attention for HFERB**
- **PTQ** strategy changes.

V. PROPOSED METHODOLOGY

The proposed methodology focuses on producing architectural improvements to parts of the previously proposed architecture. I have made sure that by approximately 5000 parameters and majorly using architectural changes to produce better results and enabling higher convergence of the base model.

A. Adaptive SRWAB: Dynamic Window Size Selection

The adaptive SRWAB enhances the original module by introducing a learnable window size mechanism and resolving implementation errors. Below are the detailed improvements. The CRAFT model leverages the Shift Rectangle Window Attention Block (SRWAB) to perform window-based attention, balancing computational cost and performance. Originally, SRWAB used fixed window sizes, limiting its adaptability to diverse image content. The introduction of an adaptive SRWAB with learnable window sizes addresses this limitation, enabling dynamic selection of window sizes to optimize feature extraction.

For this we countered challenges like:

- Dynamic window size selection using a convolutional predictor.
- Robust tensor shape handling to prevent dimension mismatches.
- Integration with the Locality Complementary Module (LCM) for enhanced feature fusion.

To adapt to different image regions, a convolutional window size predictor was added to select optimal window sizes from a predefined set (e.g., $\{[2, 4], [4, 8], [8, 16]\}$). The Simple architecture is as follows:

```
Adp. Avg Pool → Conv2d(c, 64, 3, stride = 1, padding = 1)
               → ReLU
               → Conv2d(64, |S|, 1)
               → Softmax(dim = 1)
```

where c is the input channel dimension, and $|S|$ is the number of window sizes (e.g., 3). The output is a probability map of shape $[b, |S|, h, w]$, indicating the likelihood of each window size per spatial location.

- **Selection Mechanism:** The most likely window size index is first computed per spatial location using:

$$\text{predicted_classes} = \arg \max_{\text{dim}=1} (\text{window_probs})$$

These are then flattened per sample and passed to a mode operation before finally getting the selected window size per sample.

$$\text{selected_size} = S[\text{mode_size_idx}]$$

- **Attention Modules:** Multiple `Attention_regular` modules are instantiated for each window size in $\mathcal{S}$, stored in `attns_h` and `attns_v`. For each sample in the batch, the attention module corresponding to its `mode_size_idx` is selected for horizontal and vertical attention computation.

B. Edge Aware CNN Modules

1) Motivation:

- Edges are primary components of high-frequency details in SISR.
- Max-pooling is too generic, missing subtle edge patterns critical for sharp reconstructions.

2) Implementation:

- Replaced max-pooling with a 3×3 convolutional layer initialized with Laplacian filters:

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

- Learnable weights allow adaptation to diverse edge and texture patterns during training.
- Followed by 1×1 convolution for feature refinement.

3) *Laplacian Filters:* A second-order derivative filter that highlights regions of rapid intensity change (edges):

$$\nabla^2 f(x, y) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

Such a filter is useful to detect edges and highlight the edges/corners in the images. Other methods like Canny Edges etc, can make it better however the efficiency-improvement tradeoff may not be as desirable.

C. Multi-Scale Dilated Convolutions

Dilated convolutions Expand the receptive field of a convolution by adding gaps between kernel elements.

- Modification: The high-frequency branch now includes **three parallel 3×3 dilated convolutions** with dilation rates of 1, 2, and 3.
- Combination: The **outputs** of these convolutions are **summed** to form a single multi-scale feature map, which is then refined with a 1×1 convolution.
- Integration: This **replaces** the original max-pooling and 1×1 convolution sequence.

Note: It was observed that using MSD Convolutions and Laplacian filters together didn't change the output much. One of the improvements can be selected based on experimentation and parameters.

D. Refined fusion of features (HFB)

The **Spatial Attention module** is a lightweight addition that enhances the fused features by emphasizing spatially important regions. The fused features from the cross-attention step, with shape (B, dim, H, W) , where B is the batch size, dim is the channel dimension, and H and W are the height and width (**Input**) Use simple **1×1 Convolution** Reducing

the channel dimension from dim to 1, producing a spatial attention map of shape $(B, 1, H, W)$.

Each value in this map represents the importance of the corresponding spatial location. The attention map is passed through a **sigmoid function**, scaling its values between 0 and 1. **Element-Wise Multiplication:** The attention map is multiplied element-wise with the fused features, modulating them spatially. Regions with higher attention values are amplified, while less important areas are suppressed.

The Spatial Attention mechanism aims to highlight informative regions in the spatial domain ($\text{height} \times \text{width}$) by generating an attention map. This map is applied to the input feature map in a pixel-wise manner, modulating the spatial features.

Let the input feature map be:

$$\mathbf{X} \in \mathbb{R}^{B \times C \times H \times W}$$

where B is the batch size, C the number of channels, and $H \times W$ the spatial dimensions. A 1×1 convolution is applied to reduce the channel dimension to 1:

$$\mathbf{A} = \sigma(\text{Conv}_{1 \times 1}(\mathbf{X})), \quad \mathbf{A} \in \mathbb{R}^{B \times 1 \times H \times W}$$

Here, $\text{Conv}_{1 \times 1}$ is a convolutional layer with kernel size 1×1 , and σ is the sigmoid activation function applied element-wise.

The attention map $\mathbf{A}$ is broadcast along the channel dimension and applied to the input feature map via element-wise multiplication:

$$\mathbf{X}' = \mathbf{X} \odot \mathbf{A}$$

where $\odot$ denotes element-wise multiplication, and the broadcasting ensures $\mathbf{A}$ is applied identically across all channels of $\mathbf{X}$.

Output: The output $\mathbf{X}' \in \mathbb{R}^{B \times C \times H \times W}$ retains the original shape but emphasizes the spatially significant regions of the input, enhancing the network's focus during subsequent processing.

VI. DATASET DESCRIPTION

A. DIV2K

The DIV2K (DIVERse 2K resolution) dataset is a high-quality dataset introduced for image super-resolution tasks. It consists of 1,000 high-resolution (HR) images, with 800 images used for training, 100 for validation, and 100 for testing. The images have resolutions of approximately $2K$ (e.g., 1920×1080 or higher), offering rich textures and diverse content, making it suitable for training deep learning models with strong generalisation capabilities. The dataset is often downsampled using bicubic kernels to generate low-resolution (LR) counterparts at various scale factors (e.g., $\times 2$, $\times 3$, $\times 4$).

B. Set5

Set5 is a small but widely used benchmark dataset for evaluating SISR models. It contains 5 high-quality images: *baby*, *bird*, *butterfly*, *head*, and *woman*. These images are frequently used to compare performance across different algorithms, especially in terms of PSNR and SSIM scores. Despite its limited size, Set5 is valuable due to its historical relevance and standardised usage in the literature.

C. Set14

Set14 is another popular benchmark dataset for super-resolution, containing 14 diverse images including natural scenes and man-made objects. Examples include *lena*, *pepper*, *baboon*, and *zebra*. Compared to Set5, Set14 presents more visual diversity and complexity, offering a more robust evaluation of model performance, particularly on unseen and challenging patterns.

VII. RESULTS AND ANALYSIS

The section is devoted to analyzing and discussing the results received from the experiments to examine the effectiveness and efficiency of the proposed Enhanced Efficient SISR (CRAFT) method. We report the better convergence and quality of image generation.

A. Faster Convergence and Better Stability

It was observed that applying the said changes gradually had visible effects on the quality of training that is (i) the stability of training (ii) the convergence rate of the training.

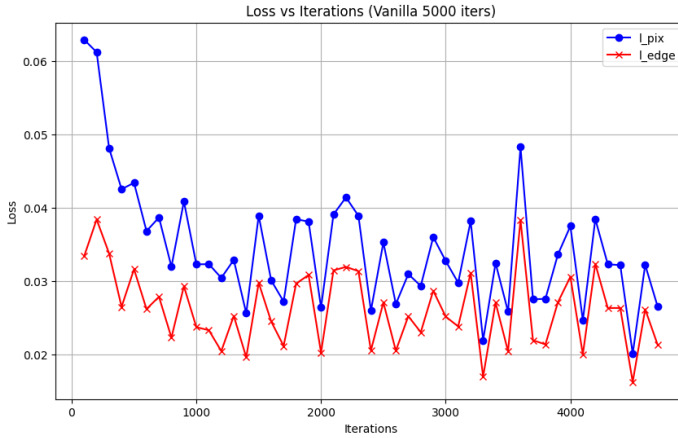


Fig. 2: Baseline CRAFT Losses 1 (pixel) and 1 (edge)

Model	PSNR	SSIM
Baseline CRAFT	26.974	0.7901
Adaptive SRWAB (Ada-SRWAB)	27.541	0.7956
Ada-SRWAB (with Laplacian Filter)	27.596	0.8015
Enhanced CRAFT	27.857	0.8135

TABLE I: Model v/s PSNR and SSIM at 5000 iterations

Fig. 2 and 3 show the final Enhanced CRAFT is more stable in training and also gives better convergence in less iterations (we have used Set5 for these plots). The final results for DIV2K for different models that we tested is given

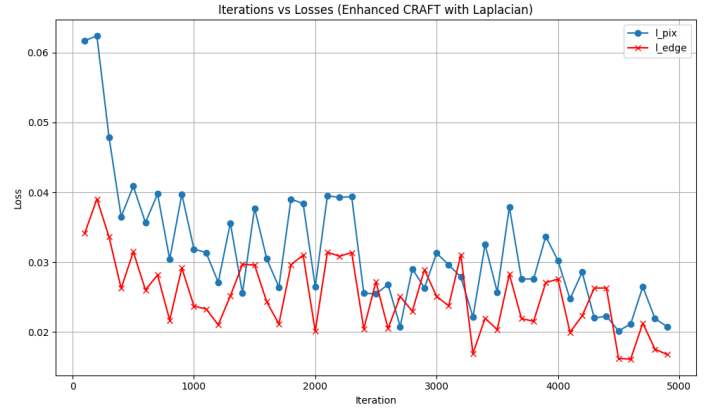


Fig. 3: Enhanced CRAFT Losses 1 (pixel) and 1 (edge)

Validation Set	PSNR	SSIM
Set5	27.857	0.8135
Set14	27.746	0.8056
BSD100	27.52	0.7218

TABLE II: Validation Sets on Enhanced CRAFT PSNR and SSIM at 5000 iterations

B. Explanation for Improvement

The three enhancements improve the SISR model by addressing key weaknesses in transformer-based architectures—particularly their tendency to lose high-frequency detail and their inefficiencies in capturing local spatial structures. Here’s a breakdown of why each improvement contributes to a better model:

1) *Enhanced SRWAB (Shifted Rectangle Window Attention Block) with Adaptive Window Size*: Traditional SRWAB uses fixed-size attention windows, which may not be optimal for all image regions—small textures might need finer granularity, while smoother areas can tolerate larger receptive fields.

By making window size adaptive, the model can dynamically choose attention scales based on the input content.

Fine textures → small windows (local context)

Smooth regions → large windows (global context)

This improves context-aware flexibility, allowing the attention mechanism to capture better dependencies that match local spatial complexity. Enhances both detail preservation and efficiency.

2) *Cross-Channel High-Frequency Enhancement (CCHFE) in HFERB using Laplacian Filtering*: Transformers are strong at global modeling but weak at local, high-frequency information (e.g., textures, edges). CNNs are better at capturing spatial detail, and the Laplacian filter is a classical edge detector that directly targets high-frequency components in the feature map. The CCHFE injects explicit high-frequency edge priors into the high-frequency branch. These priors are learnable, so the model can adapt edge emphasis based on training data.

3) *Spatial Attention after Fusion in RCRFG*: After fusing low-frequency (transformer) and high-frequency (CNN) features, some important spatial details may still be diluted. Spatial Attention re-weights the fused feature map based on where important content lies (e.g., object edges, textured regions). Generates a spatial importance map using a lightweight 1×1 convolution followed by sigmoid activation and applies this map to selectively amplify informative regions and suppress less relevant ones.

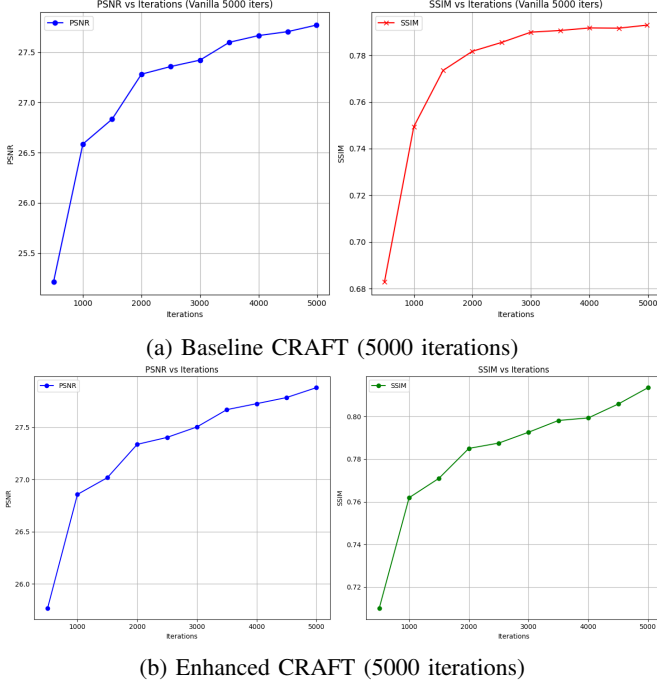


Fig. 4: Comparison of PSNR and SSIM performance between Baseline and Enhanced CRAFT models after 5000 iterations. The Enhanced CRAFT consistently demonstrates improved reconstruction quality.

VIII. CONCLUSION AND FUTURE WORK

In conclusion, the overall goal of proposing an Enhanced-CRAFT is to increase the performance of the Baseline Model using effective and most importantly, efficient changes. From the experimental results, two main claims are supported—the convergence can be easily modulated if some architectural changes are implemented, and also that the number of training steps can be reduced drastically by adding a minimal no. of parameters. Finally, the PTQ strategies discussed have applications more than just CRAFT and extend to many other approximation algorithms.

A suggested direction that we think is (i) Progressive learning and regeneration in the SRWAB block stack and (ii) Integration of less complex State Space Models, taking advantage of their breakthrough in image processing and vision.

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